The explicit twist map is a group isomorphism

Onto the entire Frobenius-minus eigenspace

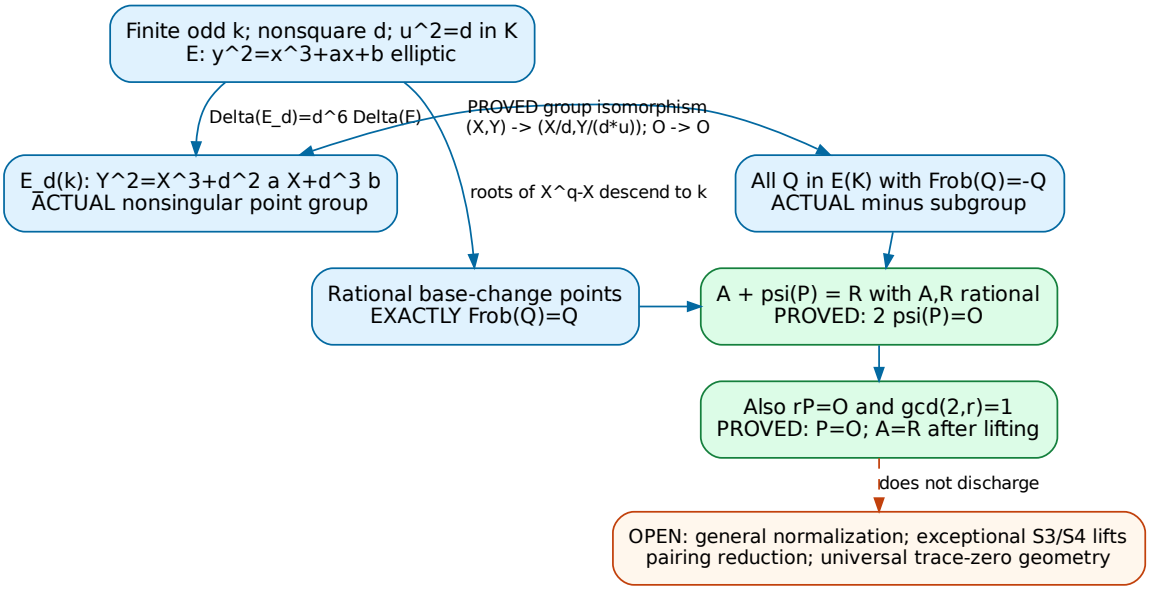
23 October 2026 · `group-map-20261023` · PR #2381

Checked result. Lean proves addition and integer-scalar compatibility of the existing actual-point quadratic-twist map, transports source annihilators, and identifies its entire minus eigenspace. Rational base-change points give exactly the fixed eigenspace. The unmodified repository proof gate passes **49 named research theorems / 57 audited declarations**, zero failing and zero extra trusted axioms. This closes a formal bridge; it is not a solver speedup, a hardness proof, or a universal no-connection theorem.

Let k be finite of odd characteristic, $q = \#k$, d a nonsquare, K/k any field algebra containing $u^2 = d$, and $E : y^2 = x^3 + ax + b$ elliptic. Set

$$E_d : Y^2 = X^3 + d^2aX + d^3b, \qquad \psi(O) = O, \quad \psi(X, Y) = (X/d, Y/(du)) .$$

There is no hidden finite-extension or degree-two hypothesis on K. The earlier proof derived $u^q = -u$ and actual Frobenius negation; its source and all old certificates remain unchanged.



group-map-20261023: actual-point Lean proofs; no solver-cost conclusion

Source-order obstruction. For rational lifts A, R, if $A + \psi(P) = R$, $rP = O$ and $\gcd(2, r) = 1$, then $P = O$ and the rational lifts are equal. The annihilator now comes from the source point through the proved group map. For a finite sum, only the aggregate is forced to zero; individual summands need not vanish. Nonzero two-torsion remains an exception when oddness is dropped.

What the checked proof actually does

Addition. The new `TwistGroupMap` module proves that the secant, tangent and vertical slopes transform by $1/u$. Addition's x and y coordinates transform by $1/d$ and $1/(du)$. The proof on Mathlib's actual nonsingular points includes O, inverse pairs and doubling; it does not assume compatibility. It constructs an actual additive homomorphism, proves

$$\psi(P+Q)=\psi(P)+\psi(Q),\quad \psi([n]P)=[n]\psi(P)\ (n\in\mathbb{Z}),\quad [r]\psi(P)=O\iff [r]P=O.$$

The group-map part needs fields, $d\neq 0$, $u^2=d$ and the elliptic short model. Finiteness, odd characteristic and nonsquareness enter the Frobenius consequences.

Full minus eigenspace. The new `TwistEigenspace` module first proves $z^q=z$ iff z comes from k, using all roots of X^q-X and the injective algebra map. The fixed field is derived, not assumed. If $Q=(x,y)$ satisfies $\mathrm{Frob}(Q)=-Q$, then $\mathrm{Frob}(x)=x$ and $\mathrm{Frob}(y)=-y$. Since $\mathrm{Frob}(u)=-u$, both dx and duy are fixed. They therefore descend to $X,Y\in k$. Substitution proves the actual equation of E_d, and

$$\Delta(E_d)=d^6\Delta(E)\neq 0$$

gives nonsingularity. The reconstructed source point maps to Q. This includes infinity and two-torsion. Combined with the proved addition and injection, `liftEquivMinus` is a genuine group isomorphism

$$E_d(k)\ \simeq\ \{Q\in E(K):\mathrm{Frob}(Q)=-Q\}.$$

The same fixed-field argument descends both coordinates of every fixed actual point on an elliptic Weierstrass model, proving exactly the rational plus-space description. This does not say every point is in either subgroup or assert a direct-sum decomposition at two-torsion.

Proof map. The canonical `lean/README.md` maps every claim to its declaration and exact hypotheses. The new modules contribute 21 named theorems: 16 group/coordinate statements and five fixed-field, discriminant and eigenspace statements. The gate also checks eight older generated helpers. Only `propext`, `Classical.choice` and `Quot.sound` occur; there are no added research axioms, placeholders, or trusted native evaluations. `lean4checker --fresh` is unavailable; no extra external replay is claimed.

Pinned sources and evidence. Lean 4.34.1 and Mathlib `d13f23b723b8a846827a245b89c10fc7d3f11612` remain pinned. Protocol commit `6fb418e8a` preceded execution. The exact new sources, all failed attempts and the passing build/axiom receipts are retained in `validation/group-map-20261023/`. Provisional axiom printouts from a failed elaboration are not proof receipts. The actual passing gate covers all five modules.

Native controls and the PR-caused catalog repair

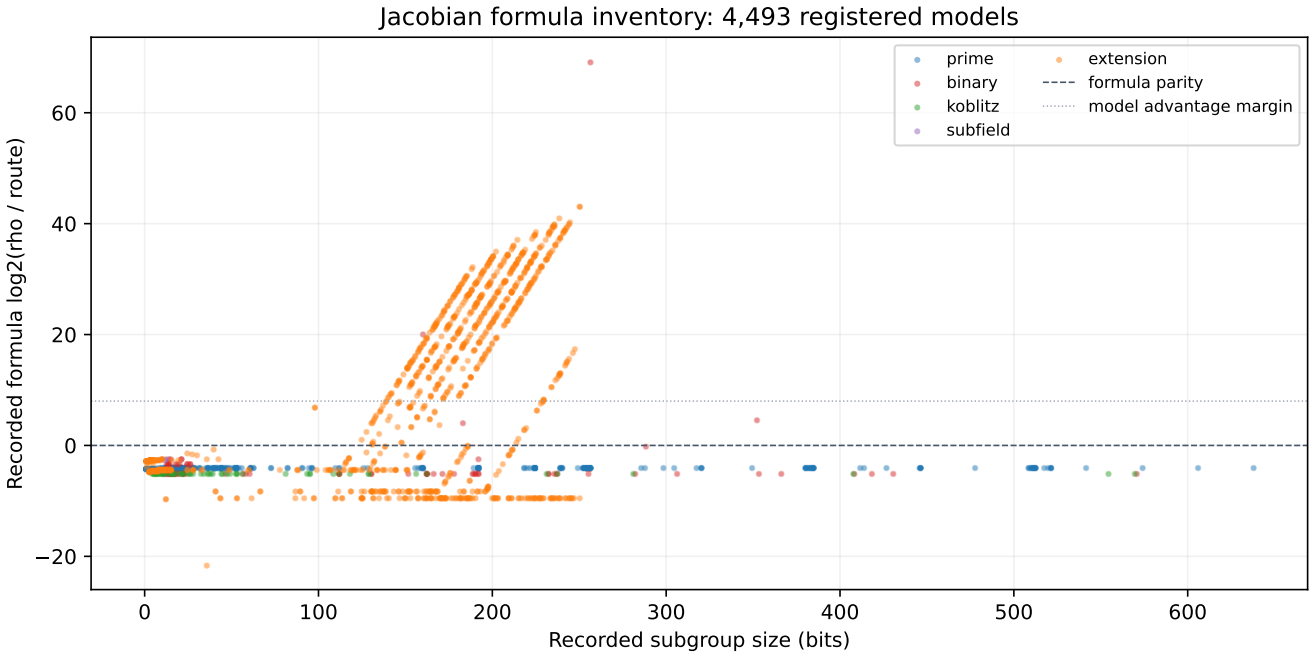
Four native IDC1 certificates bind the exact new Lean sources. Each replays 32 derived points and 16 extra points. Each false plus-one identity is rejected arithmetically with a saved counterexample, and each false source hash is rejected. The records cover polynomial parts; Lean separately checks inverses, points, addition and Frobenius.

Polynomial part	Certificate id
Tangent numerator	`IDC1h501f62832dc059fb`
Addition x coordinate	`IDC1he0c463dc2e585208`
Addition y coordinate	`IDC1hfb09521ce1a737cb`
Discriminant scaling	`IDC1hb3a3c677428bc4b1`

Metadata defect reproduced. The registry gained 1,921 models in this PR, but the Jacobian-transfer report still covered 2,572. The unchanged native generator accepts the old base registry and rejects the old report with the new registry. Under separately committed protocol `ea1929fd9`, the catalog is regenerated at its unchanged 200,000 factor budget.

Catalog field	Before	After
Models	2,572	4,493
Declared formula routes	6,565	11,365
Boundary verdicts	1,917	3,838
Formula advantage / marginal	598 / 57	598 / 57

Every prior per-model object and the cost model remain equal; exact registry membership has no missing or duplicate model. The registry itself is unchanged. Native checks, table/query/chart commands and altered-file rejection pass. The marked Markdown summary table is byte-identical after regeneration. The strict cover assessment's source binding and inventory-freshness statement are refreshed; all other assessment sections and transfer verdicts remain equal.



Recorded native values, `jacobian-transfers/2`, all 4,493 models. Axes are subgroup bits and formula $\log_2(\rho) - \log_2(\text{route})$. Points are conditional formula leads, not constructed transfers, timings, speedups or security claims. Plotting performs layout only; raw native SVG/TikZ are retained. No sampling uncertainty is asserted.

Validation, limits and next discriminating step

Requirement	Actual outcome
Study proof gate	PASS: 57 declarations, zero failing/trusted
New native certificates	Four exact-byte replays; false identities and source hashes rejected
Isolated native backend	22 tests passed, zero failed/ignored
Full release library suite	4,479 passed, zero failed, 119 existing ignores
Workflow metadata parser	Four tests passed; no Python research execution introduced
Frozen evidence/provenance	Retained; original point-map source and certificates unchanged
General mathematical request	Partial: no general hardness, advantage, GHS or no-connection proof
Shared Conductor	Still unavailable, exit 127; no reservation or replacement coordinator
Final-head CI/review/merge	Remain separate gates; PR kept open

What remains. Compatible general Weierstrass normalization group maps, exceptional S3/S4 lifting to actual points, pairing nondegeneracy and the full DLP reduction, and universal trace-zero geometry remain open in `problems/quadratic-twist-evidence-native-parity.md`. The next discriminating step is exceptional S3/S4 lifting with all exceptional roots and denominator cases retained. These are obligations, not work claimed to be running.

A connection still exists. The retained native positive example on $E/\mathbb{F}_{43} : y^2 = x^3 + x$, $d = 2$, subgroup order 11 and embedding degree two uses the twist point in the pairing and recovers the known scalar 7 by finite-field index calculus. The formal pairing-reduction bridge remains open. The present eigenspace obstruction cannot rule out every twist/IC connection. The prior 74,693,760-word finite null experiment is preserved and is not rerun or enlarged in this proof iteration. Its 942-model population differs from the 4,493-model global catalog. Invalid-curve attacks remain excluded.

Operational blockers. The starting all-event CI snapshot has 26 runs: 15 successful, four failed, four cancelled, three queued. The PR-caused stale Jacobian catalog is repaired here. The other failures are inherited SEA Lean 4.19 rejecting `--trusted`, an upstream metadata download returning 404, and CodeQL uploads reporting repository code scanning disabled. Those services and unrelated source/gates are not patched around. Complete check-runs and combined-status endpoints return 403. Reviews/threads are empty in this snapshot; a green review job is not an approval. Copilot was already asked to handle merge conflicts in browser data, traits and the leaderboard; those files are untouched. New-head CI and all completion gates must still be satisfied before merge.

Sources and review. See the source-linked `README.md` and complete `lean/README.md` proof map. Validation and raw controls are recorded in `validation/group-map-20261023/README.md`, with source/input/output digests. The editable diagram and native chart accompany every rendered copy. Earlier dated figures remain frozen; there is no measured IC/rho point or transfer promotion to add to a scoreboard.

Primary APIs: pinned Mathlib affine elliptic `Formula/Point`, finite-field `Basic`, and polynomial `Roots` modules. Mathematical background: Silverman, *The Arithmetic of Elliptic Curves*, III.1 (coordinate changes) and III.8 (pairings). Formalization status is determined by the linked declarations and receipts, not by these background citations.